

Chapter 14 Homework Answer Sheet

Part A Multiple choice (15 points)

1-5 AABCA

6-10 ECDCE

11-15 ACBBC

Part B Short questions (15 points)

16. (5 points)

- (a) (1 point) 3
(b) (2 points) 2. B and C are intermediates.
(c) (1 point) $B \rightarrow C$ step is rate determining.
(d) (1 point) For the overall reaction, ΔE is positive.

17. (5 points)

- (a) The graph of $1/[\text{NO}]$ versus time is linear with positive slope, indicating that the reaction is second order in $[\text{NO}]$. The rate law will include $[\text{NO}]^2$.
The first step is a fast equilibrium, $k_1[\text{NO}][\text{Cl}_2] = k_{-1}[\text{NOCl}_2]$.
From the rate-determining step, $\text{rate} = k_2[\text{NOCl}_2][\text{NO}] = (k_2k_1/k_{-1})[\text{NO}]^2[\text{Cl}_2]$ (2 points)
This rate law is second order in $[\text{NO}]$. It is consistent with the observed data. (1 point)
- (b) The linear plot guarantees that the overall rate law will include $[\text{NO}]^2$. Because the data were obtained at constant $[\text{Cl}_2]$, we have no information about reaction order with respect to $[\text{Cl}_2]$. (2 points)

18. (3 points)

- (a) (1 point) Half-life = 0.010 s
(b) (2 points) $t_{1/2} = 0.693/k$ $k = 0.693/0.010 \text{ s} = 69 \text{ s}^{-1}$

19. (2 points)

According to $\ln(k_2/k_1) = -E_a/R(1/T_2 - 1/T_1)$

$$k_2 = 3k_1$$

$$T_2 = 337 + 273 = 610 \text{ K}$$

$$T_1 = 327 + 273 = 600 \text{ K}$$

$$E_a = (8.314 \times \ln 3) / (1/600 - 1/610) = 3.34 \times 10^5 \text{ J/mol} = 334 \text{ kJ/mol}$$